

Probabilistic Machine Learning
(CS772A, Spring 2026)
Homework 1
Due Date: April 15, 2026 (11:59pm)

Instructions:

- Only electronic submissions will be accepted. Your main PDF writeup must be typeset in LaTeX using the provided template (please also refer to the “Additional Instructions” below) and has to be submitted via Gradescope <https://www.gradescope.com/>).

Additional Instructions

- We have provided a LaTeX template file `hw1sol.tex` to help typeset your PDF writeup. There is also a style file `pmi.sty` that contain shortcuts to many of the useful LaTeX commands for doing things such as boldfaced/calligraphic fonts for letters, various mathematical/greek symbols, etc., and others. Use of these shortcuts is recommended (but not necessary).
- Your answer to every problem should begin on a new page. The provided template is designed to do this automatically. However, if it fails to do so, use the `\clearpage` option in LaTeX before starting the answer to a new question, to *enforce* this.
- While submitting your assignment on the Gradescope website, you will have to specify on which page(s) is question 1 answered, on which page(s) is question 2 answered etc. To do this properly, first ensure that the answer to each question starts on a different page.
- Be careful to flush all your floats (figures, tables) corresponding to question n before starting the answer to question $n + 1$ otherwise, while grading, we might miss your important parts of your answers.
- Your solutions must appear in proper order in the PDF file i.e. solution to question n must be complete in the PDF file (including all plots, tables, proofs etc) before you present a solution to question $n + 1$.

Problems 1-8 (Total 160 marks; 20 marks per question)

The AI Fact-Checking Challenge: Modern Large Language Models (LLMs) are highly fluent but frequently “hallucinate” technical details, especially regarding mathematical constraints and experimental setups.

The following paper summaries were produced primarily by an AI and subsequently edited for this assignment. Each paper’s summary contains **four (or more) significant technical errors/wrong statements**.

Your Task: Your task is to identify 4 such places where the summary has flaws w.r.t. the original paper. In particular, for each paper, identify the **four errors** by quoting the incorrect phrase and providing a brief (1-2 sentence) technical correction based on your reading of the original text. **Please do also quote (but in brief) the relevant line(s)/equation(s)/figure(s) from the paper which your corrections are based on.**

Problem 1: The Paper – Variational Auto-Encoder (VAE)

Reference: Kingma & Welling (2013). “Auto-Encoding Variational Bayes.”

“The Variational Auto-Encoder (VAE) introduced the SGVB (Stochastic Gradient Variational Bayes) estimator to perform efficient inference in directed graphical models with continuous latent variables.

The architecture consists of an Encoder network that maps the input x directly to a single point vector z in a lower-dimensional latent space. To ensure the model remains end-to-end differentiable, the authors introduce the Reparameterization Trick, which involves adding uniform noise (sampled from $U[-1, 1]$) to the latent vector before it is passed to the decoder.

The model is optimized by maximizing the Evidence Lower Bound (ELBO). This objective function consists of a discriminative term used to predict the class labels of the data, and a KL Divergence term that acts as a penalty to ensure the approximate posterior $q_\phi(z|x)$ is as dissimilar as possible from the prior $p(z)$. Finally, the paper demonstrates through experiments on the Frey Faces dataset that for complex image data, the prior $p(z)$ must be a learned mixture of Gaussians to achieve high-quality reconstruction.”

Problem 2: The Paper – Black Box Variational Inference (BBVI)

Reference: Ranganath et al. (2014). “Black Box Variational Inference.”

“The Black Box Variational Inference (BBVI) framework was designed to automate variational inference. The authors propose an algorithm that only requires the user to provide the analytic gradient of the ELBO with respect to the latent variables z .

The core of the method is the Log-Derivative Trick. To ensure the algorithm converges, the authors introduce Control Variates. In this paper, control variates are used solely to bias the gradient estimates toward the global optimum, ensuring that the stochastic updates never stray into local minima. These functions must be manually derived for every new model to ensure the gradient variance is exactly zero.

To optimize the parameters, the algorithm uses a fixed, large learning rate. Finally, the paper demonstrates BBVI using Stochastic Volatility models, proving the method works even when the model’s log-likelihood $\log p(x, z)$ cannot be evaluated numerically.”

Problem 3: The Paper – Stochastic Gradient Langevin Dynamics (SGLD)

Reference: Welling & Teh (2011). “Bayesian Learning via SGLD.”

“The Stochastic Gradient Langevin Dynamics (SGLD) algorithm provides a way to perform Bayesian inference on massive datasets. The authors propose a modification to standard SGD where a fixed amount of constant variance noise is added to the gradient updates at every iteration. This noise ensures that the sampler explores the entire parameter space regardless of the step size.

Crucially, the algorithm eliminates the need for the Metropolis-Hastings (MH) acceptance step by ensuring the step size ϵ_t stays constant throughout training. This allows the chain to maintain a high rejection rate to avoid local minima. The authors prove that as long as the noise is sampled from a Uniform distribution, the distribution of the parameters θ will converge to a single MAP (Maximum A Posteriori) estimate. Finally, the paper demonstrates the method’s efficiency using a Logistic Regression model on the MNIST dataset.”

Problem 4: The Paper – Deep Neural Networks as GPs (NNGP)

Reference: Lee et al. (2018). “Deep Neural Networks as Gaussian Processes”

“The authors demonstrate that deep neural networks (DNNs) with a finite number of hidden units per layer are equivalent to Gaussian Processes. This equivalence holds because as you add more layers to a network, the Central Limit Theorem (CLT) causes the output to become normally distributed, regardless of the width of the hidden layers.

The paper introduces a recursive method to compute the Neural Tangent Kernel (NTK), which defines the covariance between inputs in a deep GP. They prove that to maintain this GP equivalence, one must use a Sigmoid activation function, as ReLU activations cause the covariance to collapse to zero at every layer. Additionally, the authors find that the GP performance is always strictly worse than the corresponding finite-width trained neural network. Finally, the researchers evaluate their theory using the MNIST dataset, showing that the GP derived from a 10-layer CNN outperforms a standard SVM.”

Problem 5: The Paper – BatchBALD

Reference: Kirsch et al. (2019). “BatchBALD: Efficient and Diverse Batch Acquisition”

“BatchBALD is an extension of the BALD objective designed for high-throughput labeling. The authors point out that standard BALD is inefficient for batch acquisition because it underestimates the uncertainty of individual points, leading the model to ignore outliers.

To solve this, BatchBALD maximizes the sum of the individual mutual informations of all points in a candidate batch. This objective ensures that the points in the batch are as similar to each other as possible to provide a ‘consensus’ update to the model parameters θ . The paper proves that this objective is linearly scalable, meaning that calculating a batch of size 1000 is no more computationally expensive than calculating a batch of size 1.

Furthermore, the authors demonstrate that BatchBALD is particularly effective for Support Vector Machines (SVMs) because they lack the ‘overconfidence’ found in Neural Networks. Finally, the

researchers evaluate their method on the Repeated MNIST dataset, showing that BatchBALD avoids the redundancy that plagues top-K BALD.”

Problem 6: The Paper – Deep Ensembles

Reference: Fort, Hu & Lakshminarayanan (2019). Deep Ensembles: A Loss Landscape Perspective

“This paper provides a theoretical justification for deep ensembles by proving that they are equivalent to exact Bayesian inference in neural networks. The authors argue that each model in the ensemble corresponds to an independent sample from the true posterior distribution over weights.

To study this phenomenon, the paper analyzes the loss landscape in weight space and shows that models obtained along a single SGD training trajectory explore multiple distinct modes of the posterior. In contrast, models initialized independently tend to converge to the same mode due to the strong convexity of the loss function.

The authors introduce a diversity–accuracy tradeoff and conclude that increasing diversity always harms predictive accuracy, making large ensembles undesirable. Finally, the experiments demonstrate that subspace sampling methods such as SWAG consistently outperform deep ensembles on out-of-distribution generalization tasks.”

Paper 7: Automatic Differentiation Variational Inference

Reference: Kucukelbir et al. (2017). “Automatic Differentiation Variational Inference.”

“Automatic Differentiation Variational Inference (ADVI) provides a general-purpose framework for variational inference by expressing the Evidence Lower Bound (ELBO) in a transformed parameter space where all latent variables follow a standard normal prior. This transformation ensures that the variational approximation can be optimized without reference to the original model constraints.

The method relies on a reparameterization of the variational distribution that eliminates the need for stochastic gradient estimates, allowing exact gradients of the ELBO to be computed via automatic differentiation. In practice, ADVI is implemented using a mean-field Gaussian approximation, as the paper shows that full-rank covariance structures offer negligible practical benefits.

A key advantage of ADVI is that it avoids the need for model-specific derivations by assuming that all latent variables are independent in the variational family, which simplifies inference without significantly affecting posterior correlations. Finally, the empirical evaluation demonstrates that ADVI consistently matches or exceeds the accuracy of Hamiltonian Monte Carlo across all tested models while being significantly faster.”

Paper 8: Laplace Redux – Effortless Bayesian Deep Learning

Reference: Daxberger et al. (2021). “Laplace Redux – Effortless Bayesian Deep Learning.”

“This paper revisits the Laplace approximation (LA) as a practical approach to Bayesian deep learning, emphasizing that its effectiveness depends primarily on the choice of curvature approximation

used to model posterior uncertainty. In particular, the authors show that different approximations (e.g., diagonal, KFAC, and full-matrix forms) lead to similar predictive performance when applied to modern neural networks.

A key contribution of the paper is demonstrating that applying LA to only the final layer provides a computationally efficient approximation that closely matches full-network Laplace across a wide range of tasks, including out-of-distribution detection. This suggests that most of the useful posterior uncertainty is captured in the last layer of the network.

The authors further argue that marginal likelihood optimization for tuning prior precision is largely unnecessary in practice, as predictive performance is relatively insensitive to this hyperparameter when using well-trained networks.

Finally, the paper concludes that Laplace approximations offer competitive uncertainty estimates compared to variational inference and deep ensembles, particularly when computational cost is taken into account, though their performance is largely invariant to the specific predictive approximation used.”